Project Development Phase: Performance Testing

Field	Details
Date	02 November 2026
Team ID	N/A (Individual Project)
Project Name	EV Value & Longevity Analytics Dashboard
Prepared by	Piyush Malviya
Maximum Marks	5 Marks

Model and dashboard performance testing record. Values were checked against figures computed independently from the CSV files in Python.

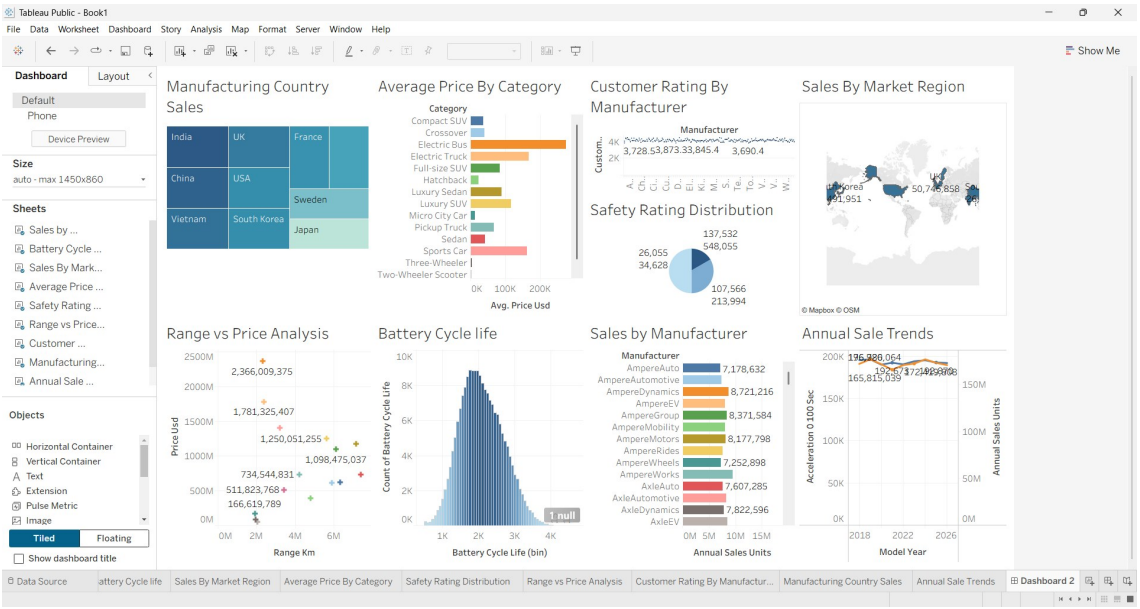
S.No.	Parameter	Screenshot / Values
1	Data Rendered	Training file: 200,000 rows x 37 columns, 1,550,703,961 annual units (matches the Tableau status bar). Test file: 100,000 rows x 37 columns, 772,608,124 units.
2	Data Preprocessing	Checked 5 columns with missing values (cycle life 1.9%, resale value 3.1%, screen size 1.6%, sales 1.0%, rating 2.0%), 1 duplicate vehicle ID, and data types. Averages ignore nulls. Price formatted in USD.
3	Utilization of Filters	Category, Battery Type, Manufacturer, Market Region and Model Year filters applied to the dashboard.
4	Calculation Fields Used	Value Index = Price USD / Range km; average price, average range, average rating and average cycle life KPIs; percent of records by safety stars.
5	Dashboard Design	No. of visualizations / graphs: 9 worksheets arranged in one dashboard (bar, histogram, map, pie, scatter, circle plot, treemap, line).
6	Story Design	No. of story points: 5 chapters (market, price, range, battery, decision).

Hold-out Validation (Train vs Test File)

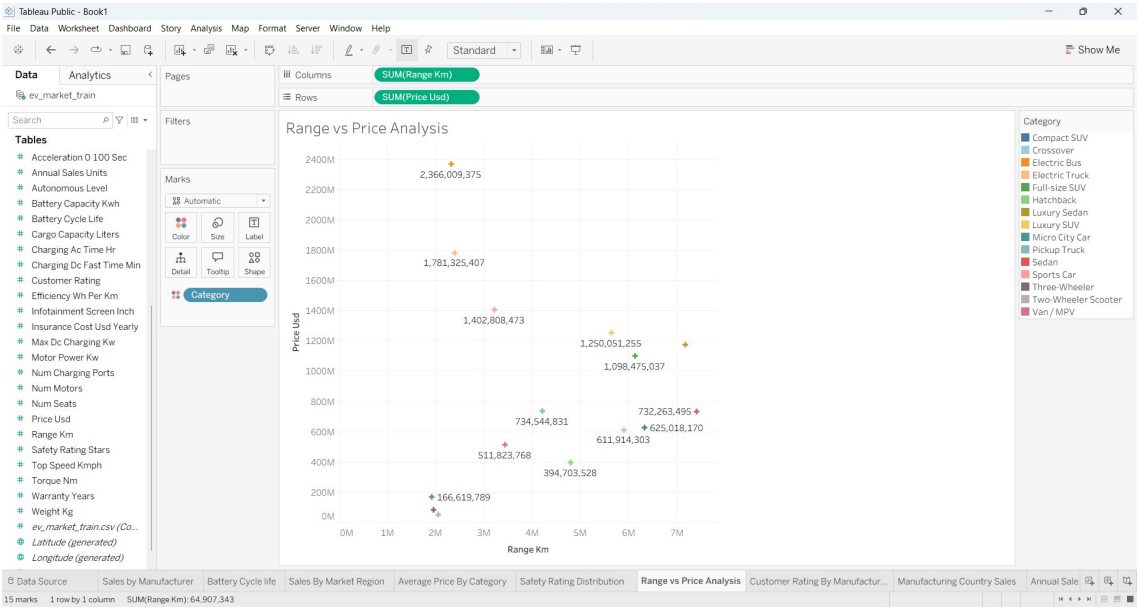
The same measures were computed on the unseen test file to confirm the dashboard logic generalises.

Measure	Train file	Test file	Difference
Average price (USD)	64,921	65,010	+0.14%
Average range (km)	324.54	324.31	-0.07%
Average customer rating	4.089	4.085	-0.10%
Manufacturers	200	200	same
Missing annual sales values	1.01%	0.99%	similar

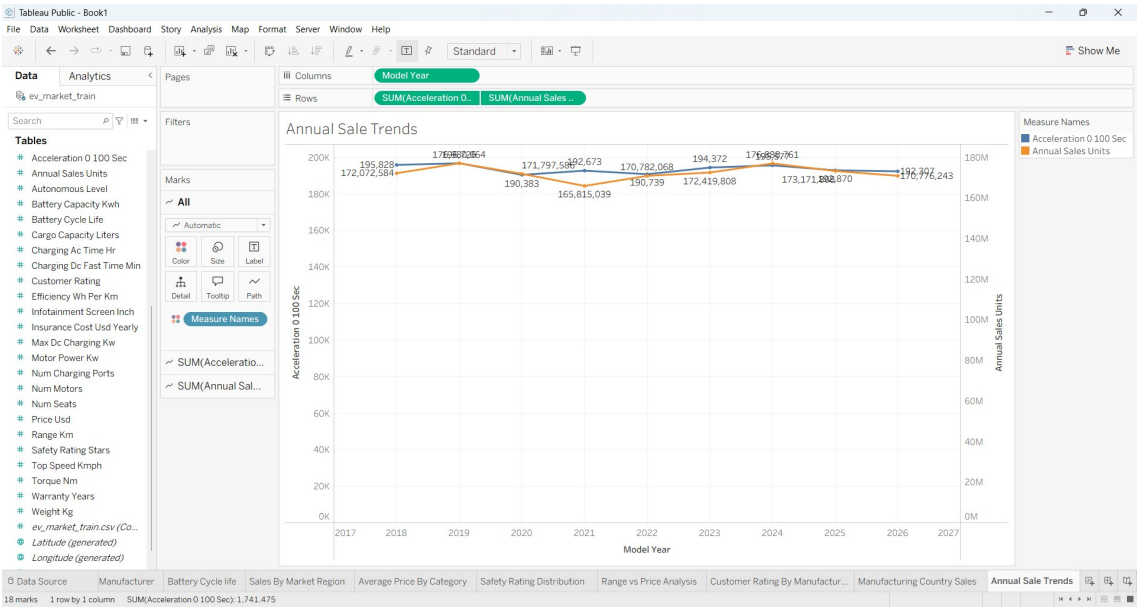
Test Evidence



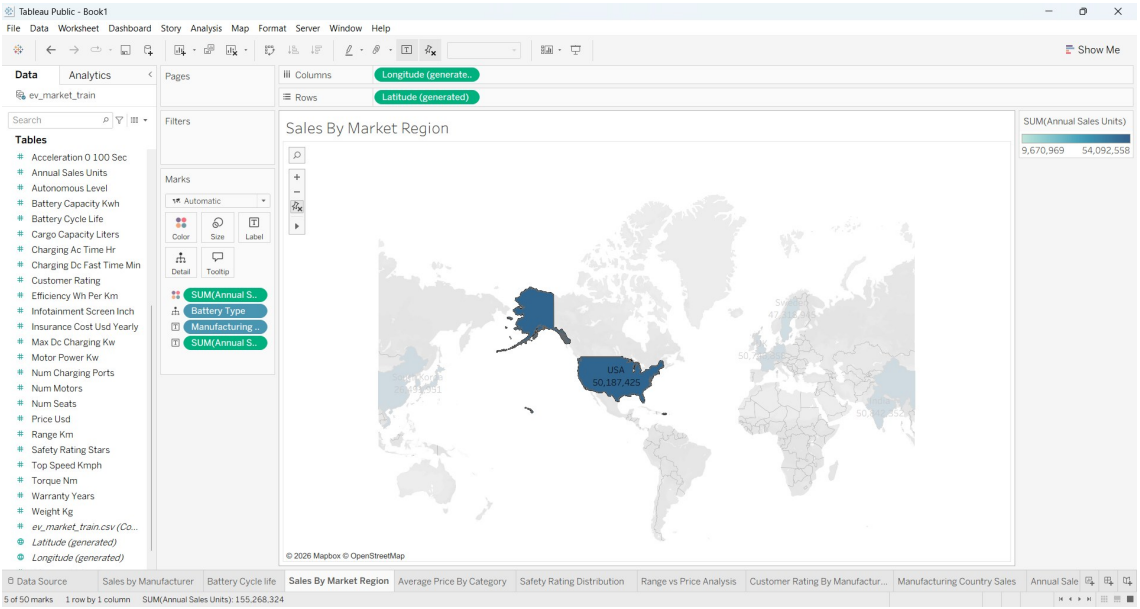
Dashboard 2 rendering all nine worksheets



Range vs Price Analysis worksheet



Annual Sale Trends worksheet



Sales by Market Region worksheet

Issues Found and Fixes

Worksheet	Issue	Fix
Customer Rating by Manufacturer	SUM used, giving values near 4,000 on a 5-star scale.	Use AVG, axis 3.9 to 4.2.
Range vs Price Analysis	SUM of both axes.	Use AVG per category with log price axis.
Annual Sale Trends	Second axis is SUM of acceleration.	Replace with average range or price.
Safety Rating Distribution	Measures unrelated to share of vehicles.	Use record count and percent labels.
Sales by Market Region	Region names are not map locations (5 of 50 marks).	Map Manufacturing Country; bar chart for region.